

DOCUMENTATION-KA MASHRUUCA DATA ANALYSIS

Student Attendance and Academic Performance Analysis

Magaca Mashruuca:	Student Attendance and Academic Analysis
Qoraaga:	Dekha Abdulkadir
Aaladda Loo Adeegsaday:	Microsoft Excel
Xogta Nadiifka ah:	297 Records (Excluding duplicates & missing values)
Geeddi-socodka (Workflow):	RAW DATA → CLEAN DATA → FORMULAS → PIVOTTABLES → CHARTS → KPIs → SLICERS → DASHBOARD → FINDINGS

1. Project Overview (Ujeeddada Mashruuca)

Purpose: Prepare, clean, analyze and visualize student academic performance data. Main variables included in the dataset are **Student_ID**, **Program**, **Attendance_Percent**, **Assignment_Score**, **Midterm_Score**, **Final_Score**, and **Academic_Status**.

The project follows the complete Excel Data Analysis workflow, focusing specifically on student attendance patterns, test scores, and overall academic standing.

2. Workbook Structure (Qaab-dhismeedka Excel-ka)

- **Original data:** Original dataset kept completely unchanged for reference (300 records).
- **Clean_Data:** Standardized and cleaned dataset ready for analysis (297 records).
- **Analysis:** Section for primary formulas, logic checks, and calculations.
- **PivotTables:** Summary tables used for aggregated analysis and generating charts.
- **Dashboard:** Interactive visual interface with key metrics, charts, and slicers.
- **Project-info / Findings:** Interpretation, core findings, and actionable recommendations.

3. Data Cleaning Process (Geeddi-socodka Nadiifinta Xogta)

Started with 300 original records and 7 variables. Through rigorous data quality audits, several issues were addressed:

- Identified 15 blank cells, 3 duplicate records, and 15 unknown/invalid values.
- Standardized Program categories (e.g., CS → Computer Science, PH → Public Health).
- Standardized Academic Status values (e.g., AR / AT RISK → At Risk, GOOD STANDING → Good Standing).
- Utilized core Excel text cleaning functions: TRIM(), CLEAN(), SUBSTITUTE(), UPPER(), and PROPER().
- Final Result: 297 high-quality cleaned records used for analysis.

4. Key Performance Indicators (KPIs)

KPI Metric	Qiimaha / Natijada
Total Cleaned Records	297
Average Attendance Rate	79.32%
Average Assignment Score	70.99
Average Midterm Score	66.45
Average Final Exam Score	71.54
Students with Final Score ≥ 70	169 Students

5. Key Findings & Recommendations (Natijoooyinka & Talo-bixinnada)

Core Findings:

- **Good Standing:** 220 students (74.07%) are performing well and hold a solid academic standing.
- **At Risk:** 55 students (18.52%) require academic intervention and structured guidance.
- **Program Lead:** Business Administration has the highest student count (59 students).
- **Attendance Lead:** Data Science achieved the highest average attendance (~82.9%).
- **Performance Lead:** Computer Science achieved the highest average final score (~72.8%).

Actionable Recommendations:

- Provide targeted academic support (tutoring, advising) for the 55 At-Risk students.
- Set up early warning attendance tracking to reach students before exam scores drop.
- Investigate remaining unknown records to ensure future data completeness.